Tutorial 9

MAT3007 Optimization

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Outline

1 KKT conditions

2 Convex Optimization by CVX

Recap: Lagrange multiplier and KKT points

1. Existence : LICQ
2. Conditions: FONC, SONC, SOSC
3. Specially important conditions : KKT conditions (and Fritz John)

Recap: Constraint Qualifications

Definition 1.

Linear Independence Constraint Qualification (LICQ) The collection of gradients

$$\{\nabla g_i(x^*) : i \in \mathcal{A}(x^*)\} \cup \{\nabla h_j(x^*) : j = 1, \ldots, p\}$$

should be linerly independent to have full rank.

Definition 2.

Regular A feasible point x^* satisfying the LICQ is called regular.

There are more CQs: e.g. ACQ, GCQ, MFCQ, PLICQ

KKT Condition

Consider a general nonlinear constrained optimization problem

$$\begin{aligned} \min \quad & f(x) \\ s.t. \quad & g_i(x) \leq 0 \quad \forall i \in 1, ..., m \\ & h_j(x) = 0 \quad \forall j \in 1, ..., p \end{aligned}$$

KKT Condition

Theorem 3.

If x is a local minimizer and a constraint qualification holds, then there exists λ and μ s.t.

1. *Main Condition*

$$\nabla f(x) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x) + \sum_{j=1}^{p} \mu_j \nabla h_j(x) = 0.$$

2. *Dual Feasibility*

$$\lambda_i \geq 0 \quad \forall i = 1, ..., m.$$

3. *Complementarity*

$$\lambda_i \cdot g_i(x) = 0 \quad \forall i = 1, ..., m.$$

4. *Primal Feasibility*

$$g_i(x) \leq 0, \quad h_j(x) = 0 \quad \forall i, \quad \forall j.$$

KKT Condition

1. Is KKT condition sufficient or necessary for being a minimizer?
   - Neither. However, if a point satisfy the constraint qualification (CQ), then KKT condition is a necessary condition for it to be a minimizer.
2. For convex problem, is KKT condition necessary or sufficient for a global minimizer?
   - Sufficient, not necessary.

KKT Condition: Remark

Proof (KKT's sufficiency in convex optimization)

Define $d := x - x^*$

$$\begin{aligned} f(x) - f(x^*) &\geq \nabla f(x^*)^T d \\ &= -\lambda^* \nabla g(x^*)^T d - \mu^* \nabla h(x^*)^T d \\ &= -\lambda^* \nabla g(x^*)^T d \geq 0 \end{aligned}$$

The second equality is due to affine-linear of g:

$$\nabla h(x^*)^T d = h(x) - h(x^*) = 0.$$

The last inequality is by the complementarity condition:

$$\lambda_i^* \nabla g_i(x^*)^T d \leq \lambda_i^* (g_i(x) - g_i(x^*)) = \lambda_i^* g_i(x) \leq 0$$

Exercise 1: Case without KKT point

Consider

$$\begin{aligned} \min\ & f(x) = x_1 + x_2 \\ s.t.\ & h_1(x) = (x_1 - 1)^2 + x_2^2 - 1 = 0 \\ & h_2(x) = (x_1 - 2)^2 + x_2^2 - 4 = 0 \end{aligned}$$

Show that the KKT condition cannot be hold for any feasible point.

Exercise 1: Solution

Firstly, we can observe that $(0,0)$ is the only feasible point.

$$\nabla h_1(0,0) = (-2,0)^T, \nabla h_2(0,0) = (-4,0)^T, \nabla f(0,0) = (1,1)^T$$

For all

$$\lambda_1, \lambda_2 \in \mathbb{R}$$

$$\nabla h_1(0,0)\lambda_1 + \nabla h_2(0,0)\lambda_1 + \nabla f(0,0) \neq 0$$

Moreover, LICQ and other CQs can not hold in this case.

Exercise 2

Consider the problem of projecting a point $v \in \mathbb{R}^n$ onto an ellipsoid

$$\begin{aligned} \min \quad & \|x - v\|^2 \\ s.t. \quad & \sum_{i=1}^{n} b_i x_i^2 \leq 1, \end{aligned}$$

where $b_i > 0$ for $i = 1, ..., n$. v is outside the ellipsoid, i.e. $\sum_{i=1}^{n} b_i v_i^2 > 1$.

1. Derive the KKT condition for the problem.
2. Describe a way to solve the problem by the KKT condition.

Exercise 2: Solution

Define $B := diag\{b_1, b_2, ..., b_n\}$.
The constraint can then be turned to $x^T Bx - 1 \leq 0$. Only if x is a non-zero vector, the gradient of constraint gets full columu rank, satisfying LICQ.
The Lagragian function is

$$\mathcal{L}(x, \lambda) = \|x - v\|^2 + \lambda(x^T Bx - 1)$$

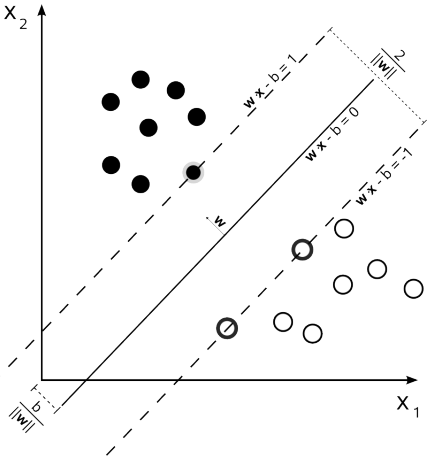
- main condition:$\nabla_x \mathcal{L}(x, \lambda) = 2(x - v) + 2\lambda Bx = 0$.
- primal feasiblity: $x^T Bx \leq 1$.
- dual feasibility: $\lambda \geq 0$.
- complementarity condition: $\lambda(x^T Bx - 1) = 0$.

Exercise 2: Solution

2. We discuss two situations:
   - $\lambda = 0$. By main condition we have $x = v$. By complementarity condition, $v^T Bv - 1 = 0$, which contradicts with $\sum_{i=1}^{n} b_i v_i^2 > 1$.
   - $\lambda > 0$. By main condition we have $x = (I + \lambda B)^{-1} v$. ($I + \lambda B$ is invertible due to its positive diagonal), we can substitute it into $x^T Bx = 1$ to get λ, then use it to calculate x.

Exercise 3

Support Vector Machine (SVM)



Exercise 3

Consider Support Vector Machine (SVM) with linear model

$$f(\mathbf{x}) = W^\top \mathbf{x} + b,$$

where W and $b_{1\times 1}$ are unkown and determined by the training samples $\{(x_i, y_i)\}_{i=1}^n$ (normally y_i equals 1 or -1). This is achieved by the following formulation:

$$\begin{aligned} \min_{W,\epsilon,b} \quad & \frac{1}{2} W^\top W + C \sum_{i=1}^{n} \epsilon_i \\ \text{s.t.} \quad & y_i \left(W^T \mathbf{x}_i + b\right) \geq 1 - \epsilon_i, && i = 1, 2, ..., n, \\ & \epsilon_i \geq 0, && i = 1, 2, ..., n, \end{aligned}$$

where C controls the trade-off between model complexity (first term) and empirical risk (second term).

- Write down the KKT conditions for SVM.

Exercise 3: Solution

Define the dual variables $\alpha \in \mathbb{R}^n$. The Lagrangian function is

$$L(W, b, \epsilon, \alpha) = \frac{1}{2} W^T W + C \sum_{i=1}^{n} \epsilon_i + \sum_{i=1}^{n} \alpha_i (y_i (W^T x_i + b) - 1 + \epsilon_i).$$

Main condition

$$W + \sum_{i=1}^{n} \alpha_i y_i x_i = 0, \quad \sum_{i=1}^{n} \alpha_i y_i = 0, \quad C + \alpha_i \geq 0, \quad i = 1..., n.$$

Primal feasibility

$$y_i (W^T x_i + b) \geq 1 - \epsilon_i, \quad i = 1, ..., n.$$

Dual feasibility

$$\alpha_i \leq 0, \quad i = 1, ..., n.$$

Complementarity

$$\alpha_i (f(x_i) - 1 + \epsilon_i) = 0, \quad i = 1, ..., n$$
$$\epsilon_i (C + \alpha_i) = 0, \quad i = 1, ..., n.$$

Outline

1 KKT conditions

2 Convex Optimization by CVX

Convex Optimization using CVX

CVX can solve a range of nonlinear optimization problems.But you need to note that:

- It can only recognize certain classes of convex functions.
- CVX only accepts models that can be proven convex using only the rules in CVX's DCP ruleset. And one may have to manually convert a problem into a recognizable form before inputting into CVX.
- here is the link of the reference:
  https://www.cse.iitb.ac.in/ cs709/notes/code/cvx/doc/dcp.html
  https://vincentqin.gitee.io/blogresource-2/cv-books/CVX.pdf

Example 1

Consider the problem:

$$\begin{aligned} \min \quad & \|Ax - b\|_2, \\ \text{s.t.} \quad & Cx = d, \\ & \|x\|_\infty \leq e \end{aligned}$$

```
1 m = 20; n = 10; p = 4;
2 A = randn(m,n); b = randn(m,1);
3 C = randn(p,n); d = randn(p,1); e = rand;
4 cvx_begin
5     variable x(n)
6     minimize( norm( A * x - b, 2 ) )
7     subject to
8     C * x == d
9     norm( x, Inf ) <= e
10 cvx_end
```

Example 2

Consider the following function:

$$\begin{array}{ll} \text{minimize} & x/y \\ \text{s.t.} & e^{-10} \leq x \leq e^3 \\ & x^2 + y/z \leq \sqrt{y} \\ & x/y = z^2 \\ & x, y, z \geq 0 \end{array}$$

Use CVX to solve it.

Solution by CVX

Let $u = \log x$, $v = \log y$, $w = \log z$, and take the logarithm of each of the constraints. The original problem is equivalent to,

$$\begin{array}{ll} \text{minimize} & u - v \\ \text{s.t.} & -10 \leq u \leq 3 \\ & \log\left(e^{2u} + e^{v-w}\right) \leq \frac{1}{2}v \\ & u - v = 2w \end{array}$$

Note that in this new problem, the first and third constraints are linear and thus are convex.

```
cvx_begin
variables u v w

minimize u-v

subject to

-10 <= u <= 3
log(exp(2*u)+exp(v-w))-0.5*v <= 0
u-v-2*w == 0

cvx_end
```

```
u = -10, v = -5, w = -2.5

x = 4.5e-5, y = 0.0067, z = 0.0821

optimal value: 0.0067
```

Figure: convex problem by cvx

Acknowledgements

This slide is based on previous tutorial materials in fall and spring semesters for MAT3007. We thank previous MAT3007 TAs and instructors for sharing previous course materials.